

DATA AVAILABILITY STATEMENT

Manuscript: Institutional Observability Under Scaled AI Governance: Deployment-Scale Capacity Degradation in Human Oversight Pipelines

Journal: Technology in Society (Elsevier)

Date: June 3, 2026

Data Availability

No primary data were collected for this theoretical and simulation-based study.

Simulation Code and Reproducibility

All simulation code, figure generation scripts, and reproducibility materials are available in a GitHub repository:

Repository: <https://github.com/SilverCoin256/oversight-saturation-sedi> **DOI (Zenodo):** [DOI to be minted upon acceptance]

The repository contains: - Monte Carlo simulation code (Python) - Kingman heavy-traffic approximation implementation - SEDI index computation scripts - Figure generation code for all five figures - Simulation parameter files - README with reproduction instructions - requirements.txt with all dependencies

Calibration Data Sources

All data used for model calibration are from publicly available sources:

1. **Vanderbilt University Center for Teaching (2023).** The Turnitin AI Detection Controversy: False Positives, Academic Integrity, and Institutional Review Capacity. Technical report. Publicly available.
2. **Peterson Health Technology Institute (2026).** The State of AI in Prior Authorization: Volume, Velocity, and Human Oversight Gaps. Technical report. Publicly available.
3. Published peer-reviewed literature as cited in the reference list.

Data Access

- Simulation outputs (CSV files of Monte Carlo runs) are included in the code repository.
- No restricted, proprietary, or confidential data were used.
- No data access restrictions apply.

Statement for Manuscript

The following statement appears in the manuscript:

“No primary data were collected for this theoretical and simulation-based study. All simulation parameters are documented in Table 1. Simulation code and figure generation scripts are available at <https://github.com/SilverCoin256/oversight-saturation-sedi>. Calibration data sources (Vanderbilt University, Peterson Health Technology Institute) are publicly available and cited in the references.”

Statement prepared by: Shaurya Gupta **Date:** June 3, 2026